

NKOLE MARTIN MWANZA

Software Engineer · ERP Developer · Full-Stack Developer

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PROFESSIONAL SUMMARY

Final-year Computer Science and Software Engineering student at UNZA with active production internship experience across ERP development, full-stack engineering, and enterprise system administration. Contributed to ERP deployments serving 3 client organisations and over 800 end users — customising 6 core modules, building 30+ custom DocTypes, and configuring automated approval workflows that reduced procurement cycle times by 40–70% and HR administrative effort by 50–80%. Currently building CCIS, a full-stack parish records management system, as a final-year capstone project. Also designs and deploys professional portfolio websites for clients, managing the complete workflow from questionnaire to GitHub Pages deployment — 3 live sites delivered to date. Comfortable across frontend, backend, ERP, and DevOps concerns, with a strong foundation in Git, GitHub, CI/CD, and version control workflows.

PROFESSIONAL EXPERIENCE

Software Engineer Intern – ERP Developer

Innovative Dynamics Ltd · Automate E ERP Platform

Nov 2025 – Present

Lusaka, Zambia (On-site)

- Contributed to ERP deployments across 2 client organisations, supporting implementations that onboarded and trained over 400 employees on the Automate E platform.
- Customised 6 core ERP modules — Procurement, Human Resources, Finance & Accounting, Maintenance, Billing, and Payroll — developing client-specific workflows, validation logic, and user interfaces beyond standard platform capabilities.
- Created and configured 30+ custom DocTypes to model new business data structures; reviewed and extended many additional existing DocTypes to meet client requirements.
- Configured end-to-end workflow automation and multi-level approval chains across all major modules, reducing procurement cycle times by an estimated 40–70% and HR administrative effort by 50–80% through elimination of paper-based processes.
- Participated in requirements analysis and ERP configuration sessions with client stakeholders across all 3 organisations, translating business needs into tested, deployable system solutions.
- Implemented server-side Python scripts and JavaScript client scripts to enforce business rules, automate notifications, and improve end-user experience across deployed modules.
- Performed REST API integrations, database configuration, data migration, and query optimisation on MySQL-backed production systems.
- Administered user accounts, role-based access controls, and permission structures across client environments, ensuring correct access segregation aligned to organisational hierarchies.
- Built custom reports, dashboards, and print formats delivering real-time financial and operational visibility — reducing management reporting preparation from days to near real-time.
- Diagnosed and resolved application issues across live deployments, providing effective technical support and improving platform stability for client operations.
- Documented system configurations, customisations, and technical processes to support knowledge transfer and ongoing platform maintainability.
- Tech: ERPNext · Frappe · Python · JavaScript · MySQL · REST APIs · Workflow Automation · Role-Based Access Control · Custom DocTypes · System Integration

EDUCATION

B.Sc. Computer Science & Software Engineering

University of Zambia (UNZA)

Expected Nov 2026

Lusaka, Zambia

- Concentrations: Artificial Intelligence & Modelling · Database Management Systems · Software Development
- Relevant coursework: Data Structures & Algorithms · Operating Systems · Computer Networks · AI · Software Engineering · Web Technologies · Database Management Systems · Computer Architecture · Statistics & Applications · Project Management
- Final-year capstone project: CCIS — Catholic Community Information System (see Projects).

PROJECTS

CCIS — Catholic Community Information System

Final-Year Capstone Project · University of Zambia · In Development

Jan 2026 – Present

github.com/Nkole101/ccis

- Designing and building a production-grade parish records management system for St Gabriel Parish — covering family management, SCC records, member profiles, and a full parish register (baptism, confirmation, marriage, communion, death).
- Architected a React 19 + Vite frontend with React Router, Recharts dashboards, Framer Motion animations, and light/dark theme support; Express.js REST API backend with PostgreSQL, bcrypt password hashing, and JWT authentication via HttpOnly cookies.
- Implemented secure authentication flows including email verification and password reset using short-lived hashed tokens; integrated Resend email service with console-link fallback for local development.
- Designed the full relational database schema across 6 core tables (SCCs, families, members, users, auth tokens, parish register entries); all schema changes managed through versioned migration files.
- Enforced role-based access control across 3 roles (superadmin, secretary, member) with route-level middleware protecting all mutations and sensitive reads.
- Maintained with structured Git version control, comprehensive README, Mermaid architecture diagrams, and full workflow and sequence documentation.
- Tech: React 19 · Vite · Node.js · Express · PostgreSQL · JWT · bcrypt · Axios · Recharts · Framer Motion · Git · GitHub

Professional Portfolio Websites

Freelance · 3 live sites delivered

2025 – Ongoing

[URLs available on request]

- Designs and builds bespoke single-page professional portfolio websites for individual clients across various professions, managing the complete end-to-end workflow: client questionnaire, content gathering, design, development, and deployment.
- Builds responsive, visually polished HTML/CSS/JavaScript portfolios covering professional summary, experience timelines, education, skills, certifications, and downloadable documents (CV, transcripts, reference letters).
- Deploys all sites to GitHub Pages, managing repositories, branch-based deployment, version control, and post-launch updates for each client.
- 3 sites currently live and accessible; additional sites in progress. Repeatable questionnaire-to-deployment workflow established for efficient client onboarding.
- Tech: HTML5 · CSS3 · JavaScript · GitHub Pages · Git · GitHub Actions · Responsive Design

Full-Stack Progressive Web Application

Personal Project

2025 – Ongoing

- Architecting a PWA using React + Vite (frontend), Node.js + Express (backend), PostgreSQL + Redis (database layer), and Nginx as API gateway. Implementing Workbox service worker caching for offline-first functionality, JWT authentication, and RESTful API design.

Procurement Planning System — Frappe ERPNext

Personal / Internship Extension Project

2025

- Built a custom procurement module with multi-role approval routing (Requestor → HOD → Finance → Director), automated email notifications, budget validation hooks, and role-based access — directly extending real internship requirements.

CERTIFICATIONS

CCNA – Introduction to Networks

Network protocols, IP addressing, routing and switching fundamentals, and network security concepts.

Cisco Networking Academy · 2025

IT Essentials – PC Hardware and Software

PC assembly, hardware configuration, software installation, troubleshooting, and preventive maintenance.

Cisco Networking Academy

NDG Linux Essentials & NDG Linux Unhatched

Cisco / NDG

Linux command line, file system navigation, user and permission management, and shell scripting fundamentals.

TECHNICAL SKILLS

Languages: Python · JavaScript · Java · SQL · Kotlin · HTML5 · CSS3 · Bash

Frontend: React 19 · Vite · React Router · Recharts · Framer Motion · Lucide React · Responsive CSS · PWA / Service Workers · Workbox

Backend: Node.js · Express.js · FastAPI · Frappe · ERPNext · REST API design · JWT · bcrypt · HttpOnly cookies

Databases: PostgreSQL · MySQL · SQLite · Redis · SQLAlchemy · data migration · query optimisation

ERP & Business Systems: ERPNext · Frappe · Custom DocTypes · Workflow Automation · Server Scripts · Client Scripts · Role-Based Access Control · ERP Customisation · Requirements Analysis

DevOps & Version Control: Git · GitHub · GitHub Pages · GitHub Actions (CI/CD) · branch-based deployment · Nginx · XAMPP

Networking & OS: TCP/IP · DNS · DHCP · routing/switching (CCNA) · Windows · Linux (Ubuntu, CLI) · shell scripting

Hardware & IT Support: PC assembly · hardware/software troubleshooting · system configuration (Cisco IT Essentials)

Tools: VS Code · PyCharm · IntelliJ IDEA · Android Studio · MySQL Workbench · ProjectLibre · Chrome DevTools · Microsoft Office Suite

Soft Skills: Team leadership · stakeholder communication · requirements analysis · technical documentation · agile/iterative development · client training

REFERENCES

Available on request.